program 6

Author

kumkum mk 1nt23is106

Write an R script to construct a box plot showcasing the distribution of a continuous variable grouped by a categorical variable using ggplot2's fill aesthetic.

Step 1: Load Required Library

```
#load ggplot2 package for visuilization
library(ggplot2)
```

step 2: Explore the Inbuilt Dataset

```
# use the built-in iris dataset
# petal width is a continuous variable
 str(iris)  #view structure of the dataset
```

```
'data.frame':   150 obs. of  5 variables:
 $ Sepal.Length: num  5.1 4.9 4.7 4.6 5 5.4 4.6 5 4.4 4.9 ...
 $ Sepal.Width : num  3.5 3 3.2 3.1 3.6 3.9 3.4 3.4 2.9 3.1 ...
 $ Petal.Length: num  1.4 1.4 1.3 1.5 1.4 1.7 1.4 1.5 1.4 1.5 ...
 $ Petal.Width : num  0.2 0.2 0.2 0.2 0.2 0.4 0.3 0.2 0.2 0.1 ...
 $ Species     : Factor w/ 3 levels "setosa","versicolor",..: 1 1 1 1 1 1 1 1 1 1 ...
```

```
 head(iris) # view sample data
```

```
  Sepal.Length Sepal.Width Petal.Length Petal.Width Species
1          5.1         3.5          1.4         0.2  setosa
2          4.9         3.0          1.4         0.2  setosa
3          4.7         3.2          1.3         0.2  setosa
4          4.6         3.1          1.5         0.2  setosa
5          5.0         3.6          1.4         0.2  setosa
6          5.4         3.9          1.7         0.4  setosa
```

Step 3: Construct Box Plot with grouping

Step 3.1: Initialize ggplot with Aesthetic Mapping

```
p <- ggplot(data = iris, aes(x = Species, y = Petal.Width, fill = Species))
p
```



Step 3.2: Add Box Plot Layer

```
p <- p + geom_boxplot()
p
```



Step 3.3: Add Labels and Theme

```
p <- p + labs(title = "Box plot of Petal Width by Species",
              x = "Species",
              y = "Petal Width") +
  theme_minimal()
p
```

